

were studying in Sections 6.1 and 6.2 were the vectors and circuits of two realizable oriented matroids of rank r respectively $n - r$,

$$\mathcal{M} = \mathcal{M}(\text{Val}(X)) \quad \text{and} \quad \mathcal{M}^* = \mathcal{M}(\text{Dep}(X)).$$

It is easy to write down extensive lists of axioms that are satisfied by any collection $\text{SIGN}(U)$. So one gets to *axiom systems* for oriented matroids. We'll get to that in Lecture 7, but without much detail. In fact, the proofs relating various axiom systems for oriented matroids tend to involve hard work, something we try to avoid on this show. (This is, however, at the basis of oriented matroid theory; we refer to [96, Ch. 3]).

The oriented matroid of a point configuration is a delicate model for its geometry. It provides a much finer model than what the *matroid* encodes about a vector configuration. (If you want to know what a matroid is, see Welsh [555], White [557], or Oxley [430].) One can prove that in fact the approximation of the combinatorial model to “geometric reality” is extremely good — this is made precise in Lawrence’s “topological representation theorem.” We'll get back to this in Lecture 7.

This means also that all the main features of the geometry of vector configurations can be derived from formal properties (the axioms). In fact there are very few geometric statements that would be true for vector configurations but fail for oriented matroids — so whatever we find in that direction is even more exciting. (See Theorem 7.20 for an example.)

(b) Equivalence

Different sets of data “ A ” and “ B ” about a geometric situation are *equivalent* if any two configurations with the same data A also have the same data B , and conversely. This means that (at least in principle) one can construct the data A from the data B . Any set of data that is equivalent to the set of circuits is referred to as the *oriented matroid* of V .

We now want to show that the four sets of data for an oriented matroid given by Definition 6.5 are equivalent. For that we need to define the combinatorial analogue of the condition $\mathbf{c}\mathbf{x} = 0$.

Definition 6.6. Let $\mathbf{x} \in \{+, -, 0\}^n$ and $\mathbf{c} \in (\{+, -, 0\}^n)^*$ be two sign vectors. Then we define that “ $\mathbf{c}\cdot\mathbf{x} = 0$ ” if

- for each i , we have $c_i = 0$ or $x_i = 0$,
- or there are indices i, j with $c_i = x_i \neq 0$ and $c_j = -x_j \neq 0$.

For a family of sign vectors $\mathcal{S} \subseteq \{+, -, 0\}^n$, we define

$$\mathcal{S}^\perp := \{\mathbf{c} \in (\{+, -, 0\}^n)^* : \mathbf{c}\cdot\mathbf{u} = 0 \text{ for all } \mathbf{u} \in \mathcal{S}\},$$

and analogously for collections of row vectors.

In fact, the condition $\mathbf{c} \cdot \mathbf{x} = 0$ holds if and only if there are real vectors $\mathbf{x} \in \mathbb{R}^n$ and $\mathbf{c} \in (\mathbb{R}^n)^*$ such that $\text{sign}(\mathbf{x}) = \mathbf{x}$, $\text{sign}(\mathbf{c}) = \mathbf{c}$, and $\mathbf{c}\mathbf{x} = 0$. For example, we have

$$(+00-)\begin{pmatrix} + \\ 0 \\ - \\ + \end{pmatrix} = 0, \quad \text{but} \quad (+00+)\begin{pmatrix} + \\ 0 \\ - \\ + \end{pmatrix} \neq 0.$$

The reader might want to check the next two statements for the small 3-vector configuration after Definition 6.5. They are both solid theorems in the setting of oriented matroids [96, Ch. 3]. For the realizable case, we won't have a lot of problems with them.

Lemma 6.7. *Let $U \subseteq \mathbb{R}^n$ be a vector subspace of dimension r , and let $\mathbf{u} \in U$ be a vector with $\text{sign}(\mathbf{u}) = \mathbf{u} \in \text{SIGN}(U) \subseteq \{+, -, 0\}^n$.*

The vector $\mathbf{u}$ can be written as a finite sum $\mathbf{u} = \mathbf{u}_1 + \dots + \mathbf{u}_k$ of $k \leq r$ vectors $\mathbf{u}_i \in U$ whose sign vectors $\mathbf{u}_i := \text{sign}(\mathbf{u}_i)$ are below $\mathbf{u}$ and minimal, that is, such that $\mathbf{u}_i \leq \mathbf{u}$, and $\mathbf{u}_i \in \text{MIN}(\text{SIGN}(U))$.

Proof. The vectors $\mathbf{v} \in U$ whose sign vector $\text{sign}(\mathbf{v})$ is componentwise smaller than or equal to $\text{sign}(\mathbf{u})$ form a polyhedral cone:

$$C(\mathbf{u}) := \{\mathbf{v} \in U : \text{sign}(\mathbf{v}) \leq \text{sign}(\mathbf{u})\} \subseteq U.$$

This cone is in fact pointed: all vectors $\mathbf{x} \in C(\mathbf{u})$ satisfy $\sum_{i=1}^n u_i x_i \geq 0$, with equality only for $\mathbf{x} = \mathbf{0}$.

Thus $P(\mathbf{u}) := \{\mathbf{x} \in C(\mathbf{u}) : \sum_{i=1}^n u_i x_i = 1\}$ is a polytope of dimension at most $r - 1$. By the results in Lecture 1 (with Carathéodory's Theorem 1.15) every point in $P(\mathbf{u})$ can be written as a convex combination of at most r vertices. By linearizing we get that $\mathbf{u}$ is the sum of $k \leq r$ vectors on extreme rays (1-faces) of $C(\mathbf{u})$. Finally, we observe that the sign vectors on the proper faces of $C(\mathbf{u})$ are strictly smaller than $\mathbf{u}$, and thus the minimal nonzero sign vectors are precisely found on the extreme rays. $\square$

Proposition 6.8. *For any vector subspace $U \subseteq \mathbb{R}^n$ we have*

$$(\text{MIN}(\text{SIGN}(U)))^\perp = (\text{SIGN}(U))^\perp = \text{SIGN}(U^\perp).$$

Proof. Let $\mathbf{u} \in \text{MIN}(\text{SIGN}(U))$ and $\mathbf{c} \in \text{SIGN}(U^\perp)$. Then we can find $\mathbf{u} \in U$ and $\mathbf{c} \in U^\perp$ with $\text{sign}(\mathbf{u}) = \mathbf{u}$ and $\text{sign}(\mathbf{c}) = \mathbf{c}$. Since $\mathbf{c}\mathbf{u} = 0$, we get $\mathbf{c} \cdot \mathbf{u} = 0$ by definition. This implies

$$(\text{MIN}(\text{SIGN}(U)))^\perp \supseteq (\text{SIGN}(U))^\perp \supseteq \text{SIGN}(U^\perp).$$

For the converse, let $\mathbf{c} \in \{+, -, 0\}^n \setminus \text{SIGN}(U^\perp)$. Then the conditions

$$\mathbf{c} \in U^\perp, \quad \text{sign}(\mathbf{c}) = \mathbf{c}$$

— a system consisting of linear equations and strict inequalities — have no solution. Now we use a Farkas lemma, for example as follows.

Write U in the form $U = \text{lin}\{\mathbf{v}_1, \dots, \mathbf{v}_r\} =: \text{lin}(V)$, and define index sets $Z := \{i : c_i = 0\}$, $P := \{i : c_i > 0\}$, and $N := \{i : c_i < 0\}$. With this, $\mathbf{c} \notin \text{SIGN}(U^\perp)$ says that the system of inequalities and equalities

$$\begin{aligned} d\mathbf{v}_k &= 0 & \text{for } 1 \leq k \leq r \\ d_i &> 0 & \text{for } i \in P \\ d_i &< 0 & \text{for } i \in N \\ d_i &= 0 & \text{for } i \in Z \end{aligned}$$

has no solution $\mathbf{d} \in (\mathbb{R}^n)^*$. Since every positive multiple of a solution $\mathbf{d}$ for this system would be a solution as well, we get that equivalently the following system has no solution:

$$\begin{aligned} d\mathbf{v}_k &= 0 & \text{for } 1 \leq k \leq r \\ d_i &\geq +1 & \text{for } i \in P \\ d_i &\leq -1 & \text{for } i \in N \\ d_i &= 0 & \text{for } i \in Z. \end{aligned}$$

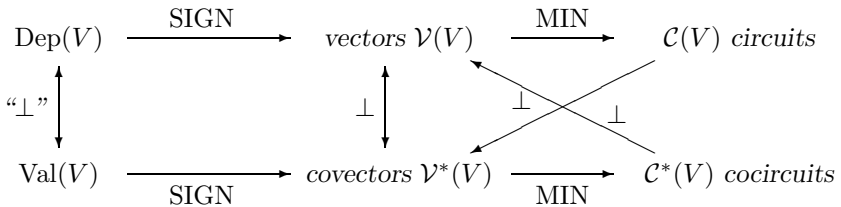
Now we apply the Farkas lemma (Proposition 1.7 adapted for systems with inequalities and equations, see Exercise 1.6) to get existence of

$$\begin{aligned} \mathbf{x} = \begin{pmatrix} \mathbf{x}' \\ \mathbf{x}'' \end{pmatrix} \in \mathbb{R}^{n+r} : \quad & x''_i \geq 0 \text{ for } i \in N \\ & x''_i \leq 0 \text{ for } i \in P \\ V\mathbf{x}' + \sum_{i=1}^n \mathbf{e}_i x''_i = \mathbf{0} \quad \text{and} \quad & \sum_{i \in P} x''_i - \sum_{i \in N} x''_i < 0. \end{aligned}$$

Now letting $\mathbf{u} := V\mathbf{x}'$, we get that $\mathbf{u} \in U$ with $\mathbf{c} \cdot \text{sign}(\mathbf{u}) \neq 0$ — in fact, we get “ $\mathbf{c} \cdot \text{sign}(\mathbf{u}) > 0$ ” in the obvious sense.

This proves that $\mathbf{c} \notin (\text{SIGN}(U))^\perp$. Furthermore, if we decompose $\mathbf{u} = \mathbf{u}_1 + \dots + \mathbf{u}_k$ into minimal vectors according to Lemma 6.7, then we find that $\mathbf{u}_i \cdot \mathbf{c} \neq 0$ has to hold for some i , and this yields a certificate to see that $\mathbf{c} \notin (\text{MIN}(\text{SIGN}(U)))^\perp$. $\square$

Corollary 6.9. *For any vector configuration $V \in \mathbb{R}^{r \times n}$, the four sets of data given by Definition 6.5 determine each other (denoted by “ $\longrightarrow$ ”), as follows:*



Thus any of the four sets of data determines the other three, and thus also the oriented matroid $\mathcal{M}(V)$.

Proof. We get this by applying Proposition 6.8 both to $\text{Dep}(V)$ and to $\text{Val}(V)$, which are dual to each other by Proposition 6.4. $\square$

Therefore two (labeled) configurations of vectors or points have *the same oriented matroid* if they have the same set of circuits, or (equivalently) the same set of cocircuits, the same set of vectors, or the same set of covectors.

(c) Duality

A concept of *duality* is built into the whole structure of oriented matroids. In fact, since the vectors and the covectors of an oriented matroid $\mathcal{M}(V)$ arise in the same way as the sign vectors of a subspace, they also satisfy the same axioms (ignoring a switch from row vectors to column vectors).

Definition 6.10. The *dual* of an oriented matroid $\mathcal{M}$ is the oriented matroid $\mathcal{M}^*$ with the following properties:

- The vectors of $\mathcal{M}^*$ are the covectors of $\mathcal{M}$, and thus the circuits of $\mathcal{M}^*$ are the cocircuits of $\mathcal{M}$.
- The covectors of $\mathcal{M}^*$ are the vectors of $\mathcal{M}$, and thus the cocircuits of $\mathcal{M}^*$ are the circuits of $\mathcal{M}$.

From the way we have defined (realizable) oriented matroids, it is clear that for every oriented matroid $\mathcal{M}$, there is a unique dual oriented matroid $\mathcal{M}^*$, whose dual is

$$(\mathcal{M}^*)^* = \mathcal{M}.$$

In fact, $\mathcal{M} = \mathcal{M}(U)$ is realizable with $U \subseteq \mathbb{R}^n$, then $\mathcal{M}^* = \mathcal{M}(U^\perp)$, for the orthogonal space

$$U^\perp := \{\mathbf{c} \in (\mathbb{R}^n)^* : \mathbf{c}\mathbf{x} = 0 \text{ for all } \mathbf{x} \in U\}.$$

So existence and uniqueness of the dual follow from Proposition 6.4 (in the realizable case):

$$\mathcal{M}^* = \mathcal{M}(\text{Dep}(V)) \quad \text{for} \quad \mathcal{M} := \mathcal{M}(\text{Val}(V)).$$

Furthermore, if $\mathcal{M}$ has rank r , then $\mathcal{M}^*$ has rank $n - r$, and conversely.

(d) Deletion and Contraction

There are two very natural and fundamental operations on point configurations: *deletion* and *contraction*. They are dual to each other, and they directly translate into oriented matroid language and terminology.